YISHUN INNOVA JUNIOR COLLEGE
H2 Geography
Paper 1

Suggested Answer

Section A

Structured Questions

Answer both questions in this Section

Cluster 1 : Development, Economy and Environment

1 Resource 1 shows e-waste collection rate in the world. Resource 2 shows the reliance on the use of fossil fuels. Resource 3 shows the growth of renewable energy between 2013 and 2022. Resource 4 shows the variations in the growth of specific renewable energy in 2022.

*E-waste is electrical and electronic equipment of any kind that has been discarded. It includes practically anything powered by an electrical source (e.g. from a power socket or a battery). Items like desktop, laptop and tablet computers, mobile phones, TVs, refrigerators, air conditioners, washing machines, rice cookers, microwave and toaster ovens, electric kettles, food processors and blenders, electric fans, DVD/video/music players, radios, hi-fis, vacuum cleaners, lamps and lighting devices, batteries, electronic toys, sports and leisure equipment, etc.

(a) Cite date from Resource 1 to describe the e-waste collection rate in the world. [3]

- Highest Europe and Northern America at 46.9 and Lowest Latin America and Carrib at 11.9
- Eastern and SEA 2nd position, but 35 points away, huge gap. Seems like the World average is at 22.8 due to Europe and N America
- Eventhough Australia and NZ can be regarded as developed regions of the world, the e-waste collection is lower than that of Eastern and SEA at 9.0
- Overall developing countries seem to have a lower score for e-waste collection eg: African nations and Latin and Caribbean

(b) With reference to Resource 1, suggest ways to curb e-waste in the different parts of the world. [5]

- **Promote Product Longevity**: Encourage manufacturers to design products with longer lifespans by making them more durable, repairable, and upgradable. This can include providing easy access to spare parts, offering software updates, and creating modular designs that allow users to replace or upgrade individual components rather than the entire product.
- **Implement Extended Producer Responsibility** : Governments can enforce regulations that require manufacturers to take responsibility for the entire lifecycle of their products, including the take-back, recycling, and safe disposal of electronic goods. This policy incentivizes companies to design more sustainable products and reduce e-waste.
- **Encourage Consumer Awareness and Education**: Educating consumers about the environmental impact of e-waste and the importance of responsible disposal can lead to more conscious purchasing decisions and better recycling practices. Public awareness campaigns can promote the reuse, repair, and recycling of electronics.
- **Establish E-Waste Collection and Recycling Infrastructure**: Develop and expand accessible e-waste collection points and recycling centers, especially in regions where such facilities are lacking. Governments and private sectors can collabourate to create efficient systems for the safe disposal and recycling of electronic waste.
- **Promote the Circular Economy**: Encourage the adoption and education of a circular economy model in the electronics industry, where products are designed for reuse, refurbishment, and recycling. This approach minimizes waste by keeping materials in use for as long as possible and reducing the need for new raw materials.

(c) Despite efforts to reduce carbon emissions, there is still reliance on the use of fossil fuels. Suggest reasons for this trend using Resource 2. [5]

- **Established Infrastructure**: Fossil fuels have been the dominant energy source for over a century, leading to a well-established global infrastructure for extraction, transportation, refining, and distribution. This extensive network makes fossil fuels easily accessible and cost-effective compared to newer, less established energy sources.
- **Energy Density and Reliability**: Fossil fuels, especially oil and natural gas, have a high energy density, meaning they can store and produce a large amount of energy per unit. This makes them highly reliable for powering industries, transportation, and electricity generation, especially in regions where renewable energy technologies are less developed or unreliable.
- **Economic Factors**: The fossil fuel industry is a significant contributor to the global economy, providing jobs, revenue, and economic stability in many countries. Transitioning away from fossil fuels can be costly and disruptive, particularly in economies that heavily depend on fossil fuel extraction and export.
- **Political Influence and Subsidies**: Fossil fuel companies have substantial political influence and often receive significant government subsidies, which help keep fossil fuel prices artificially low. This financial support creates a barrier to the widespread adoption of renewable energy sources, which may require initial investments and higher costs.
- **Transition Challenges**: Moving to renewable energy sources requires large-scale changes in technology, infrastructure, and behavior. The transition can be slow due to technical challenges, the need for innovation, and the time required to develop and scale up renewable energy solutions to meet global demand.

(d) With reference to Resource 3, explain for the spatial and temporal variation of the usage of renewable energy. [6]

- **Global Growth in Renewable Energy Capacity**:
  - The graph shows a consistent increase in renewable energy capacity globally from 2013 to 2021. This trend reflects the growing awareness of the need to transition away from fossil fuels to mitigate climate change, improve energy security, and promote sustainable development.
  - The total global renewable energy capacity increased significantly, driven by technological advancements, policy incentives, and falling costs of renewable energy technologies such as solar and wind power.
- **Regional Variations**:
  - **Asia (Other Asia & Oceania, China)**: Asia, particularly China, has seen the most substantial growth in renewable energy capacity. China's capacity increased by +223%, making it a global leader in renewable energy investments and deployment. This growth is due to China's aggressive policies to reduce air pollution, enhance energy security, and dominate the global clean energy market. China used to be the workshop of the world, manufacturing was a leading sector and it may have paved the way for China to become more conservative in the use of Fossil Fuels.
  - **Europe and North America**: These regions have also shown significant growth, with Europe increasing its capacity by +79% and North America by +65%. Europe has been at the forefront of the renewable energy transition due to strong regulatory frameworks, financial incentives, and public support for clean energy. These are also DCs which are main members of Supranational Organisations which march for the use of Renewable Energy use.
  - **South/Central America & Caribbean**: This region saw a +145% increase in capacity, driven by investments in hydroelectric, wind, and solar projects, particularly in Brazil and other countries with favorable renewable energy conditions. It could be the location of these places which favour HEP eg: Brazil and Amazon, and Sun or Wind Power eg: Caribbean Islands.
  - **Middle East & Africa**: The growth in renewable energy capacity in the Middle East & Africa (+66%) is slower compared to other regions. However, there has been a notable increase, particularly in solar power, as countries in this region seek to diversify their energy sources and reduce reliance on fossil fuels. It is lower due to the lack of financial sources, lack of government intervention or stability in the region.
- **Temporal Trends**:
  - The graph illustrates that renewable energy capacity has increased steadily each year, reflecting the growing global commitment to transitioning to clean energy sources.

- The increase in capacity is more pronounced in the later years (2018-2021), indicating accelerating investments and deployments of renewable energy technologies as costs have fallen and the urgency to address climate change has risen.

- **Challenges and Disparities**:
  - The graph notes that international clean energy aid to developing countries decreased by 3.5% from 2012 to 2021. This decline in aid may have contributed to the slower growth in renewable energy capacity in regions like Africa and the Middle East, where financial and technical barriers are more significant.
  - Despite global growth, there are disparities in renewable energy capacity development across regions, influenced by factors such as economic resources, policy support, access to technology, and regional energy needs.

(e) Explain the trade-offs associated with the use of renewable energy shown in Resource 4. [5]

- **Impact on Wildlife and Ecosystems**:
  - *Wind Energy*: Wind turbines can pose threats to birds and bats through collisions. The construction of wind farms may also disrupt local habitats.
  - *Hydroelectric Power*: Dams can alter aquatic ecosystems, affect fish migration patterns, and change the natural flow of rivers, impacting downstream ecosystems.
  - *Solar Energy*: Large-scale solar farms require significant land use, which can lead to habitat loss and ecosystem disruption if not properly managed.
- **Resource Extraction and Material Use**:
  - The production of renewable energy technologies requires the extraction of minerals and rare earth metals, which can lead to environmental degradation, pollution, and social conflicts if not conducted responsibly.
  - Manufacturing processes for solar panels and batteries can produce hazardous waste and consume significant amounts of water and energy.
- **Intermittent Energy Supply**:
  - *Solar and Wind Energy*: These sources are weather-dependent and can produce variable outputs, leading to challenges in meeting consistent energy demand.
  - **Grid Stability Issues**: The fluctuation in energy supply can cause instability in power grids, requiring additional measures to balance supply and demand.
- **Need for Energy Storage and Backup Systems**:
  - To address intermittency, substantial investments are needed in energy storage technologies like batteries or alternative backup systems, which can be costly and technologically challenging.
  - Developing efficient and large-scale storage solutions is critical but remains a significant hurdle for widespread renewable integration.
- **High Initial Investment**:
  - The upfront costs for installing renewable energy infrastructure can be substantial, posing financial barriers, especially for developing countries and low-income communities.
  - Funding and financing renewable projects can be challenging without adequate government support, incentives, or access to capital.
- **Cost Competitiveness**:

- While costs for renewable technologies have been decreasing, in some regions, they may still not be as cost-competitive as traditional energy sources without subsidies or policy support.
- The economic feasibility can vary greatly depending on local conditions, availability of resources, and market dynamics.

(f) With reference to all Resources, explain if sustainable development efforts by countries have been successful. [6]

1. **Growth in Renewable Energy Capacity (2013-2021)**:
    - The significant increase in renewable energy capacity globally, particularly in regions like China, Europe, and North America, is a positive indicator of progress in sustainable development. The expansion of renewable energy suggests that many countries are making strides in reducing their reliance on fossil fuels, thereby lowering their greenhouse gas emissions.
    - China, for example, has seen a 223% increase in renewable energy capacity, driven by large-scale investments in solar and wind power. Similarly, Europe's 79% increase highlights its commitment to transitioning to a cleaner energy mix.
2. **Variation in Fossil Fuel Use (2022)**:
    - Despite the growth in renewable energy, the reliance on fossil fuels remains high in many countries, particularly those that are rapidly industrializing or have abundant fossil fuel resources. The image showing the share of energy consumption by fuel type in 2022 indicates that countries like South Africa, China, and India still depend heavily on coal for energy, with fossil fuels accounting for more than 82% of primary energy use globally.
    - This reliance on fossil fuels, especially coal, is a significant barrier to achieving sustainable development goals, as it contributes to environmental degradation and climate change.
3. **Renewable Electricity Generation by Source (2020)**:
    - The chart depicting the breakdown of renewable electricity generation by source in 2020 shows that hydropower is the dominant renewable energy source, accounting for 59% of renewable electricity. While hydropower is a clean energy source, it comes with trade-offs such as environmental impacts on river ecosystems and communities.
    - The diversification of renewable energy sources, with wind and solar contributing 21% and 11% respectively, is a positive sign. However, the relatively small share of bioenergy (8%) and geothermal (1%) indicates that more effort is needed to tap into a broader range of renewable resources.
4. **Challenges and Disparities**:
    - The global increase in renewable energy capacity is uneven, with regions like the Middle East & Africa and South/Central America & Caribbean experiencing slower growth compared to Asia and Europe. This disparity suggests that sustainable development efforts are more successful in some regions than others, often

due to differences in financial resources, technological capabilities, and policy support.

- The decline in international clean energy aid to developing countries by 3.5% from 2012 to 2021 highlights a challenge in global sustainable development efforts. Developing countries, which are often most vulnerable to the impacts of climate change, may struggle to invest in renewable energy without sufficient financial and technical support.

Conclusion:

Overall, sustainable development efforts have been partially successful, with notable achievements in the expansion of renewable energy capacity and diversification of energy sources. However, the continued heavy reliance on fossil fuels in many countries and the disparities in renewable energy adoption across regions indicate that much work remains to be done.

Success in sustainable development is uneven and depends on various factors, including economic resources, technological access, policy frameworks, and international cooperation. For a truly sustainable future, these efforts must be intensified and made more inclusive, particularly in regions where progress has been slower. This involves not only increasing investments in renewable energy but also addressing the environmental, social, and economic trade-offs associated with different

Cluster 2 : Tropical Environments

2 Resource 5 shows the climate graph of Dhaka, Bangladesh. Resource 6 is a photograph of a part of the Brahmaputra River in the middle course of the river. Resource 7A and 7B show photographs of Dhaka during a flood. Resource 8 shows the map of the Ganges Brahmaputra Delta in Bangladesh.

(a) Cite data from Resource 5 to describe the temperature and rainfall characteristics of Dhaka, Bangladesh. [4]

- It experiences a distinct wet and dry climate.
- The Köppen-Geiger climate classification is Aw (if students say Am, we can accept, as long as they say A climate).
- The mean temperature prevailing in the city of Dhaka is recorded as 25.3 °C.
- The precipitation level on a yearly basis amount to 2055 mm.
(the above the most important descriptions)

(b) Sketch and annotate the features of the Brahmaputra river shown in Resource 6. [4]



- Flanking channels
- Braid bars
- Different colours show different sizes of load
- Mid-Channel bar
- (diagram must be the same as shown if not deduct 2m)

(c) Account for the formation of the Brahmaputra River using Resources 5 and 6. [6]

- Res 6 shows that the river is a braided one

- Res 5 shows the condition needed for the formation of a braided channel which is fluctuating discharge
- The rainfall shows a distinct wet and dry climate, rainfall comes in from May to September about 1500mm of rainfall, more than 80% falls during this period after which there is a dry climate, rainfall reduces and then is negligible from Nov to Feb.
- This type of climate directly affects the discharge in the Brahmaputra. During periods of high rainfall there is higher erosion and braid bars may be submerged, but during low rainfall periods the braid bars will be formed and reappear showing characteristics of a braided channel.

(d) With reference to Resources 7A and 7B, explain the challenges of implementing appropriate flood management strategies in Dhaka, Bangladesh. [6]

- Too many people, Res 7A, on one side developed, neat, high-rise buildings, perhaps businesses also, city area, On the other side, clustered slums, where to evacuate them to, rural to urban migration. Too much of resettlement required
- What about financial strains, the city may be dependent on the people shown in Res 7A, the high -rise buildings. Eradicating that will be a major economic strain for the city
- Slums hold the power, elections, politics
- Perhaps to have levees, most suited, to widen, straighten and deepen the river, but for that there will be construction in the region and some people or a major part fo the people living near the river will have to be resettled.
- Electricity lines, shophouses, lack of drainage, which means that just one type of flood management may not be enough, perhaps drains, but where to construct, tey have to create outlets for the water, where, under the shop houses, beside te shophouses, also becos of the type of houses seen, dirty, drains will clog up
- A problem seen in mature and overcrowded cities of the world such as Dhaka

(e) Describe the characteristics of the Ganges Brahamaputra Delta shown in Resource 8. [4]

- Drained by many rivers – Hoogly, Kobadak, Sibsa
- Subaerial above the sea, Sundarbans and Subaqueous part, below sea
- The older part is the Moribund delta then there is at the most active delta which is neared to the Bay of Bengal
- 2 types of active areas of the Delta, Tidally Active and Active Delta
- Tidally Active is still being moulded by the tides from the sea while the Active Delta is the part which is actively receiving the sediment from the rivers

(f) Using Resource 8, explain the impact of human activities on the ecosystem services provided by the Ganges Brahmaputra Delta. [6]

- Dams – HEP, reduce flooding, sediments reduced which is a threat to Deltas, lack of fertility, reduces the volume of sediment and water, delta will retreat
- Diversion of river, shrink the size of the delta
- Human population is high, unsustainable living , degradation, pollution, affect the soil and water
- Quality of water and soil deteriorates
- Rising sea level due to climate change – susceptible to flooding, wash away large volumes of sediments into the sea that form the delta, destroy livelihood of people

Section B

Answer **Either** Question 3 **or** Question 4 and
Either Question 5 **or** Question 6.

Cluster 1 : Development, Economy and Environment

3 "Extractive industries have shaped economies, societies, and politics of nations—for good and bad." [20]

Evaluate the success of managing extractive industries in low income countries.

Indicative content

Managing extractive industries in low-income countries presents both opportunities and challenges. While the potential for economic development and revenue generation is significant, the success of managing these industries is often mixed, influenced by factors such as governance, infrastructure, environmental sustainability, and social impact. In the low income countries, the failures very often far outweighs the benefits seen in high income countries due to the poor government regulation and heavy dependence on expat labour and foreign machineries,

Here's an evaluation of the success of managing extractive industries in low-income countries:

1. Economic Contribution

- **Positive Impact:** Extractive industries, such as mining and oil extraction, can contribute significantly to national GDP, create jobs, and provide essential infrastructure investments in low-income countries. Countries like Botswana and Chile have leveraged their natural resources to achieve economic growth and improve living standards.
- **Challenges:** However, many low-income countries struggle with the "resource curse," where reliance on extractive industries leads to economic volatility, inflation, and underinvestment in other sectors. Additionally, the benefits often fail to reach the broader population, leading to **increased inequality**. (e.g., In Zimbabwe, les than 1% of the population is engaged in the diamond sector, even though diamond export from Zimbabwe is about $400 million a year).

2. Governance and Corruption

- **Effective Management:** Some countries have established institutions like sovereign wealth funds to manage resource revenues, aiming to ensure long-term economic stability and development. Norway's model is often cited as successful, though it is in a high-income context. This shows that the low income countries have less effective management tools on the extractive industries.

- **Weak Governance:** In many low-income countries, weak governance and corruption undermine the effective management of extractive industries. Misallocation of resources, embezzlement, and lack of transparency often result in lost revenues and minimal development impact. Examples include Nigeria and Angola, where oil wealth has led to significant corruption and underdevelopment.

3. Environmental Impact

- **Environmental Management:** Proper environmental management and regulations can mitigate the negative impacts of extractive industries, such as deforestation, water contamination, and loss of biodiversity. Successful cases include countries that have implemented strict environmental regulations and rehabilitation programs. Nigerian oil spill messes created by Shell have ruined farms and fisheries and the Nigerian are demanding for compensation before Shell moved out (https://www.climatechangenews.com/2024/02/13/shell-accused-of-trying-to-wash-hands-of-nigerian-oil-spill-mess/)
- **Environmental Degradation:** In many low-income countries, environmental regulations are poorly enforced or non-existent, leading to severe environmental degradation. The Amazon rainforest in Brazil, for instance, has suffered from extensive deforestation due to illegal mining activities, contributing to global environmental concerns.

4. Social Impact

- **Positive Social Outcomes:** When managed effectively, extractive industries can lead to improved social outcomes, such as better education, healthcare, and infrastructure. Some countries have used resource revenues to fund social programs and reduce poverty. In Nigeria, Shell, an active supporter of STEM education, has programmes that support those who teach maths to help improve numeracy levels in beneficiary schools, and their annual bursary fund provides high performing pupils with the opportunity to pursue STEM topics.
- **Social Conflicts and Displacement:** However, extractive industries often lead to social conflicts, displacement of communities, and human rights abuses. The Democratic Republic of Congo (DRC), rich in minerals like cobalt, has experienced severe social unrest and exploitation, with local communities often bearing the brunt of environmental and social costs.

5. Long-term Sustainability

- **Sustainable Development:** The long-term success of managing extractive industries depends on sustainable practices that ensure resource revenues contribute to lasting development. Diversifying the economy, investing in human capital, and promoting environmental sustainability are key factors.
- **Resource Depletion:** Many low-income countries face challenges in ensuring long-term sustainability. Over-reliance on extractive industries can lead to resource depletion without sufficient reinvestment in other economic sectors, leaving countries vulnerable when resources are exhausted.
- **Need to evaluate that low-**income countries are reliant on the developed countries for imported technology to develop their extractive industries, and are more concerned with short-term gains than long-

term sustainability issues, hence the benefits are also not sustainable in the longer term.

Conclusion

The success of managing extractive industries in low-income countries is mixed and highly context-dependent. While these industries offer significant economic opportunities, challenges related to governance, corruption, environmental degradation, and social impact often undermine their potential. Success requires strong governance, transparent institutions, sustainable practices, and a commitment to using resource revenues for long-term development. Without these, the risks of the resource curse, social conflict, and environmental harm remain high.

4 "Population is a strong driver of the economy as well as the quality of the labour force." [20]

Evaluate the influence of labour on economic activities.

Indicative content

One approach to this essay is to see labour as a factor of economic activities. And evaluate the role of labour vis-à-vis other critical factors such as role of technology, role of government. Interestingly, when looking at these other factors, therein lies the role for the labour as well. For instance, with skill intensity and technology, the labour is not seen in term so the headcount, but can be used to intensify productivity. Government besides the industrial and investment policy can also through the education policy intensify the productivity of labour. In addition, labour is not seen as static but can move and will be attracted to where the income is more attractive.

Labour is a critical factor influencing economic activities, shaping the productivity, growth, and sustainability of economies. The availability, quality, and cost of labour impact various sectors, from agriculture to manufacturing and services. The success of the economic activities depends on the ability of tap on the labour to maximise its potential.

1. Labour Availability and Economic Growth

- **Positive Influence:** A large and available labour force can drive economic growth by meeting the demands of expanding industries. Countries with a high population growth rate, like India, often benefit from a "demographic dividend," where a growing workforce supports increased production and consumption.
- **Challenges:** However, if the labour force grows faster than job creation, it can lead to high unemployment or underemployment, limiting economic growth. Hence the economy can only benefit from the huge labour supply if it knows how to tap on the manpower. Inability to supply sufficient jobs to the population will only lead to overpopulation. On the other hand, a similarly large population canbe seen as underpopulated if the labour supply cannot meet the demands of the economy. Additionally, a shrinking labour force, as seen in aging populations like Japan, can constrain economic expansion by reducing the available workforce.

2. Labour Quality and Productivity

- **Skilled Labour:** The quality of labour, determined by education, training, and experience, directly affects productivity. High-skilled labour is essential for industries that rely on technology and innovation, such as information technology, biotechnology, and advanced manufacturing. Economies that invest in education and skills training, like South Korea, often experience higher productivity and economic growth.
- **Low-Skilled Labour:** Conversely, low-skilled labour is more prevalent in developing economies, where industries such as agriculture, mining, and basic manufacturing dominate. While these sectors are crucial for

economic development, their lower productivity can limit overall economic growth and competitiveness in the global market.

- **Hence only investment in the labour will reap the equivalent benefits.** Give egs of countries which invest in education, skill trainings to illustrate.

3. Labour Costs and Competitiveness

- **Low Labour Costs:** In some economies, low labour costs attract foreign investment and drive economic activities in labour-intensive industries like textiles, electronics, and assembly manufacturing. Countries like Bangladesh and Vietnam have benefited from their competitive labour costs, becoming major exporters in these sectors.
- **High Labour Costs:** On the other hand, high labour costs can reduce competitiveness, particularly in industries where wages are a significant portion of production costs. High labour costs may drive companies to automate processes or relocate to regions with cheaper labour, potentially leading to job losses in high-cost economies. However, high labour costs can also lead to higher standards of living and a strong domestic market, as seen in countries like Germany and Sweden. In addition, only with high wage will the best talents to attracted to boost the growth of the economy. ('brain drain' Vs 'brain gain')

4. Labour Market Flexibility

- **Flexible Labour Markets:** A flexible labour market, where workers can easily move between jobs and industries, can enhance economic resilience and adaptability. This flexibility is crucial in economies undergoing rapid technological change or facing economic shocks. The United States, for example, benefits from a relatively flexible labour market that supports innovation and entrepreneurship. This is where you can show that the labour employed in the TNCs may be lacking in geographical mobility and can be susceptible to the 'footloose nature of the TNCs'. Once the TNCs shifted out of a country, the low skilled labour may become redundant and unemployable as their skills are non-transferable)
- **Rigid Labour Markets:** In contrast, rigid labour markets with strong labour protections, high employment security, and inflexible wage structures can hinder economic activities by reducing the ability of businesses to adjust to changing economic conditions. While these protections can provide stability for workers, they may also discourage investment and limit job creation.

5. Globalisation and Labour

- **Global Labour Markets:** Globalisation has expanded labour markets, allowing companies to outsource labour-intensive activities to regions with lower labour costs. This has influenced economic activities by creating global supply chains and shifting manufacturing and service jobs across borders. However, it has also led to job displacement and wage pressure in high-income countries.
- **Labour Mobility:** The movement of labour across borders, whether through immigration or the global movement of skilled professionals, has had a significant impact on economic activities. Migrant labour fills gaps in industries facing shortages and can drive growth in host

economies, while also contributing to remittances that support economies in workers' home countries.

Conclusion

Labour is a fundamental influence on economic activities, affecting productivity, competitiveness, and overall economic growth. The success of an economy in leveraging its labour force depends on factors such as education and skill levels, labour costs, market flexibility, and supportive policies. While labour can drive economic success, challenges such as unemployment, low productivity, and the impact of globalisation must be managed to ensure sustainable and inclusive growth.

Cluster 2 : Tropical Environments

5 ‘Monsoons play a principal role in characterising the climate of the tropics.’ [20]

Evaluate this statement with reference to the climate in the tropics.

Indicative content

Monsoons are indeed crucial to the climates of tropical regions, significantly shaping weather patterns for Am which is having a distinct wet and dry climate. The effects of the monsoons, however, are reduced for the Af and Aw climates. They also do not affect the climates of the BSh and BWh climates. The effects of the monsoons vary across the 5 different climates.

Argument 1 : **The location of the Am climates are crucial in them being affected by the monsoons (spatial)**. Am climate is found in coastal regions within the tropics, particularly in areas influenced by monsoon winds. Examples include parts of South and Southeast Asia. These regions are often situated near large bodies of water, which play a crucial role in the development of monsoon systems.

The defining feature of the Am climate is the monsoon circulation, which results from the differential heating of land and sea. During the summer, the land heats up more rapidly than the ocean, creating low-pressure areas that draw in moist air from the sea, leading to heavy rainfall. In the winter, the reverse occurs, with cooler land temperatures leading to the dominance of dry offshore winds, which cause the dry season.

Give example of Either Cochin, Mangalore or Akyab, Myanmar to talk about the effects of monsoons – Asian Monsoons

Argument 2: **Monsoon effects not stable every year (temporal).**- But the strength and timing of the monsoon can vary from year to year, leading to fluctuations in the amount of rainfall received. This variability can impact agriculture, water resources, and overall ecosystem health in regions with an Am climate. Talk about failure of monsoons, lesser rainfall than expected, failure of crops

Climate change poses a threat to the stability of monsoon patterns, which could lead to more extreme weather events or shifts in the timing of wet and dry seasons.

Another climate that is affected by the monsoons is the Aw climate. The Aw is influenced greatly by the ITCZ which brings in rainfall. It coincides with the monsoons. The monsoons move from a region of HP to a region of LP that coincides with the ITCZ. But the **location of the Aw regions do not allow it to receive the full effects of the Monsoons.**

Aw regions are typically located between the equatorial zones (with Af climates) and the subtropical zones. They lie farther from the equator than regions with an Am (tropical monsoon) climate, meaning they are on the fringes of the areas most directly affected by the monsoon circulation. As a result, the monsoon winds that reach these regions are often less intense and bring less moisture compared to areas closer to the equator.

Regions with Aw climates are often located further inland or in areas where the topography and distance from large water bodies limit the amount of moisture-laden air that can reach them. Coastal regions, particularly those closer to the equator, tend to receive more intense and sustained monsoon rains.

Argument 3 : Location - In some regions, particularly those on the fringes of the Af climate zone, monsoons can contribute to the overall rainfall. For example, in coastal areas where the Af climate borders an Am climate, monsoons can enhance rainfall during certain times of the year. Eg: Singapore, Iquitos, Peru

However, even in these areas, the influence of monsoons does not lead to a pronounced dry season, as the **Af climate's defining characteristic is the absence of a dry period.**

As seen so far the monsoon are crucial in characterising the climate of the humid tropics. However, the effects vary across the 3 climates. The full effects of the monsoons are mostly felt by the Am climates with the Af and Aw receiving just a portion of the effects due to their location.

Argument 4 : Dry climates – The monsoon effects are not felt by the dry climates BSh and BWh which are also part of the Tropics. These climates are defined by their aridity, with limited and highly variable rainfall, and are influenced by factors that generally suppress the effects of monsoons.

These climates are mostly affected by the STHPB, Rainshadow effects, Cold Ocean Currents. These factors impede even mechanisms for uplift.

The whole argument is about the location of the different climates (Space) and how the location shows the variation in the effect of the monsoons.

6 ‘Physical weathering processes are common in higher latitudes while chemical weathering processes in the lower latitudes. [20]

Evaluate this statement.

Indicative content

Note that the rates of physical weathering and chemical weathering differ by latitudes (as suggested Peltier). In addition, van't Hoff's principle suggests that the rate of chemical reactions, including chemical weathering, increases with temperature. Specifically, his rule states that for every 10°C rise in temperature, the rate of a chemical reaction roughly doubles. This implies that in warmer climates, chemical weathering processes, such as the dissolution of minerals or hydrolysis, will occur more rapidly. Thus, temperature plays a key role in determining the intensity and speed of chemical weathering in different environments.
Also bearing in mind that that the depth of weathering differs between PW and CW (Strakhov).
So,
PW – temperature fluctuations are crucial with little rainfall but for CW – high rainfall with lesser temperature fluctuations are crucial.

Content 1
PW common in higher latitudes – Temperate regions, STHPB areas where there is a diurnal change in temperature. Peltier theory can be explained. However, there are exceptions to this rule.

Freeze Thaw Action – Cold regions, fluctuations around 0 degrees Celsius (can describe a little bit of the weathering process). Its also about the constant repeat of the process that affects the breakdown of the rocks through FTA. **But not necessary only the higher latitudes, mountainous regions can also experience FTA process.** (Argument 1)

Then Thermal Fracture – Dry regions, fluctuations, high diurnal temperature ranges, very cold in the night, can reduce to even negative temperature and can be very hot in the day time eg: 45 degrees Celsius, expansion and contraction. Usually deserts are common in the STHPB regions of the world – **but there are also deserts in the tropics – continental ones and also those near coastal regions, cold currents or due to topographic barriers. (Argument 2)**

Salt Weathering – Require high evaporation rates, **usually near the coastal regions**, not necessarily higher latitudes, arid regions, with low rainfall but high temperatures (Argument 3)

Content 2
CW common in lower latitudes – high steady temperatures and high rainfall, Peltier, rapid or strong CW processes of Oxidation, Carbonation, Hydrolysis. Lower latitudes have the necessary mechanism for uplift, rainfall for chemical reactions. The temperature is also high and almost constant so CW reactions

are faster. Van't Hoff rule to be explained here. Also lower latitudes, high rainfall, bring about conditions for vegetation growth – Biological processes to take place.

Also explain Carbonation process and Oxidation process.

But does **not mean that CW is absent in the higher latitudes the rates will be much lower** (Argument 4) due to the lower rainfall

As for BP – vegetation, growth, **prise open the rocks**, which is a PW process (Not in the higher latitudes) **but in the lower latitudes**. (Argument 5)

Content 3
Some weathering processes do not require high or low latitudes eg: Pressure release – its more about mass movement, erosion where the bedrock is exposed resulting in exfoliation or onion peeling. Its is about the pressure (Argument 6)

[Total : 100 marks]